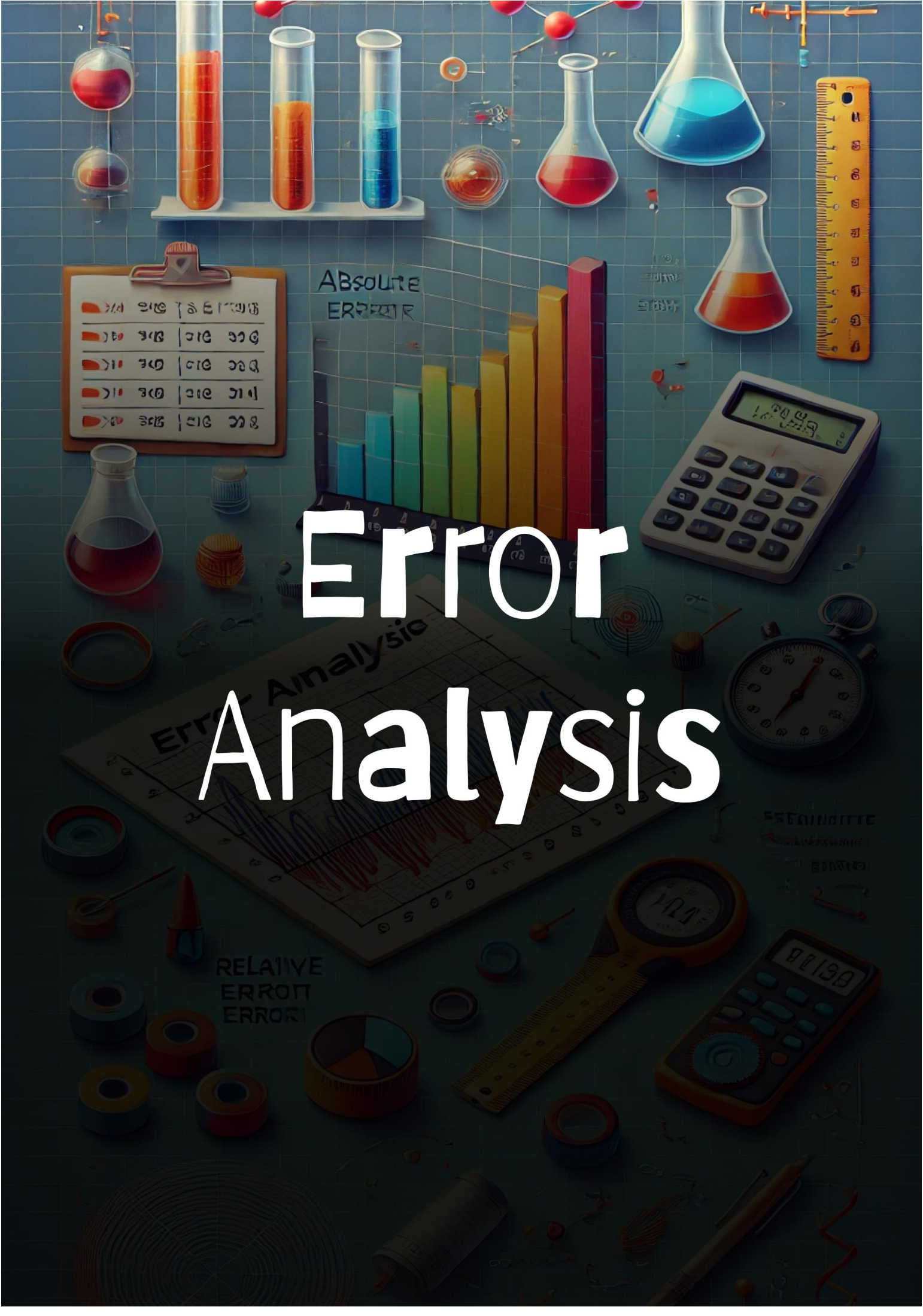

Error
Analysis

Error Analysis

PROPAGATION OF ERRORS

IN ADDITION

Let two physical quantities be given as

$p_1 = P_1 \pm \Delta P_1$

& $p_2 = P_2 \pm \Delta P_2$, then

in the sum $(p_1 + p_2)$, the errors get added

i.e. $P_1 + P_2 = P_1 + P_2 \pm (\Delta P_1 + \Delta P_2)$

★ all operations should be done as per rules of significant fig.

Eg: $R_1 = 10.0 \pm 0.1\,\Omega$

$R_2 = 10.7 \pm 0.1\,\Omega$

$R_{eff} = 20.7 \pm 0.2\,\Omega$

IN SUBTRACTION

Let two quantities be given as

$p_1 = P_1 \pm \Delta P_1$

$p_2 = P_2 \pm \Delta P_2$, then

their difference $(p_1 - p_2)$ is given by

$p_1 - p_2 = P_1 - P_2 \pm (\Delta P_1 + \Delta P_2)$

IN MULTIPLICATION

Case I: When relative errors are small

Relative error in product is sum of relative errors of terms being multiplied.

$p_1 = P_1 \pm \Delta P_1$ where $\Delta P_1 << P_1$

$p_2 = P_2 \pm \Delta P_2$ where $\Delta P_2 << P_2$

$q = p_1 p_2$

$\ln q = \ln p_1 + \ln p_2$

$$\frac{dq}{q} = \frac{dp_1}{p_1} + \frac{dp_2}{p_2}$$

$$\frac{\Delta q}{q} \simeq \frac{\Delta P_1}{P_1} + \frac{\Delta P_2}{P_2}$$

$$p_1 p_2 = q \pm \Delta q$$

Case II : When relative errors are large.

(Say >10%) then we need to physically find out the upper and lower bound of the product & report the answer accordingly.

Eg - $V = 10V \pm 10\%$

$I = 5A \pm 10\%$

$P_{up} = V_u I_u = (1.1)(5.5) = 60.5W$

$P_{Lo} = V_L P_L = (9)(4.5) = 40.5W$

$P = 50.5 \pm 10\ W$

IN DIVISION

Relative errors get added if relative errors are small.

$$q = \frac{p_1}{p_2}$$

$$\ln q = \ln p_1 - \ln p_2$$

$$\frac{dq}{q} = \frac{dp_1}{p_1} - \frac{dp_2}{p_2}$$

$$\frac{\Delta q}{q} \simeq \frac{\Delta p_1}{p_1} + \frac{\Delta p_2}{p_2}$$

However, in case relative errors are large, we will once again need to find upper & lower bounds.

Eg - $V = 10V \pm 10\%$

$I = 5A \pm 10\%$

$R = 2\Omega \pm 20\%$

$= 2 \pm 0.4\Omega$

$R_U = \frac{11}{4.5}$, $R_L = \frac{9}{5.5}$

$$R = \frac{1}{2}\left(\frac{11}{4.5} + \frac{9}{5.5}\right) \pm \frac{1}{2}\left(\frac{11}{4.5} - \frac{9}{5.5}\right)$$

$$= 2.035 \pm 9.405\,\Omega$$

IN MULTIPLICATION OF POWERS / SURDS EXPRESSIONS

Here, we estimate the max relative error by taking log & differentiating.

Que.) Find relative error in T in terms of relative error in ℓ and g.

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

$$\ln(T) = \frac{1}{2}\ln\ell - \frac{1}{2}\ln g$$

$$\frac{dT}{T} = \frac{d\ell}{2\ell} - \frac{dg}{2g}$$

$$\left|\frac{dT}{T}\right|_{max} = \frac{1}{2}\left|\frac{d\ell}{\ell}\right| + \frac{1}{2}\left|\frac{dg}{g}\right|$$

$$\frac{\Delta T}{T} \lesssim \frac{1}{2}\frac{\Delta\ell}{\ell} + \frac{1}{2}\frac{\Delta g}{g}$$

Que.) $P = \frac{A^a}{B^b}$

$$\frac{\Delta P}{P} \lesssim a\frac{\Delta A}{A} + b\frac{\Delta B}{B}$$

Que.) length of pendulum = 1 ± 0.1 m
& $g = 9.8 \pm 0.1\ m/s^2$
Find T of a pendulum. $\left(T_0 = 2\pi\sqrt{\frac{1}{9.8}}\right)$

$$\frac{\Delta T}{T} = \frac{1}{2}\left(\frac{0.1}{1}\right) + \frac{1}{2}\left(\frac{0.1}{9.8}\right)$$

$$\frac{\Delta T}{T} = \frac{1}{2}(0.1)\left[1 + \frac{1}{9.8}\right]$$

$$= \frac{0.1}{2}\left[\frac{9.9}{9.8}\right]$$

$$\frac{\Delta T}{T} \times 100 = \frac{1}{2}(10) + \frac{1}{2}(1) \lesssim 5.5$$

$T = 2.077 \pm 5.5\% \rightarrow 0.11039$

$T \simeq 2.0 \pm 0.1$

IN ARBITRARY FUNCTIONS

There is no full proof method for finding errors in general functions as some functions are so sensitive that a little change in one quantity can lead to an enormous change in the value of the function.

However, for most well-behaved function, with reasonably small errors in their parameters, we can find the error in the function as

If $W = f(x, y, z)$

then $\Delta W \simeq \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y + \frac{\partial f}{\partial z} \Delta z$

Eg: $y = \sin(x \pm \Delta x)$

$y = \sin x \pm \frac{\partial}{\partial x} \sin(x) \cdot \Delta x$

$y = \sin x \pm \cos(x) \cdot \Delta x$

$y = f(x \pm \delta x)$

then $y = (\pm \delta x \, f'(x)) + f(x)$

Eg: $\sin 32^\circ = 0.5299$

$= \frac{1}{2} + \frac{\sqrt{3}}{2} \left(2 \cdot \frac{\pi}{180}\right)_{90}$

$= \frac{1}{2} \left(1 + \frac{\sqrt{3}\pi}{90}\right)$

$= 0.53023$

DIMENSIONS

Fundamental quantities which can be used to express all other physical quantities.

CGS units	Fundamental Dimensions	SI Unit
g.	Mass	Kg.
cm.	length	m.
Sec.	Time	Sec.
K.	$\theta \rightarrow$ Temp.	Kelvin
Cd.	Cd $\rightarrow$ Candela (luminous Intensity)	Candela
	A $\rightarrow$ Current	ampere
	n $\rightarrow$ moles / quantity of matter	

- Velocity LT^{-1}
- Acceleration LT^{-2}
- Force MLT^{-2}
- Energy ML^2T^{-2}
- Current A
- Momentum MLT^{-1}
- Power ML^2T^{-3}
- Planck's Constant ML^2T^{-1} $\left(\frac{E\lambda}{c} = \frac{ML^2T^{-2}.L}{LT^{-1}}\right)$

Que.) Find dimensions:

→ Electric Potential → $ML^2T^{-3}A^{-1}$

→ Electric Field → $MLT^{-3}A^{-1}$

→ Resistance → $ML^2T^{-3}A^{-2}$

→ μ_0 → $\left(\frac{q}{EA}\right)\left(\frac{AT}{MLT^{-3}A^{-1}L^2}\right)$

$= M^1L^{-3}A^2T^4$

→ Inductance → $\left(\frac{\text{Energy}}{i^2}\right) = ML^2T^{-2}A^{-2}$

→ Coefficient of viscosity (η) $\left(\frac{f}{\alpha v}\right)\left(\frac{ML T^{-2}}{L L T^{-1}}\right) = ML^{-1}T^{-1}$

Que.) If force (F), velocity (v), acceleration (a) were to be used as fundamental dimensions, what will be dimensions of M, L, T.

$$\boxed{M} = F^x v^y a^z$$

$$M = (MLT^{-2})^x (LT^{-1})^y (LT^{-2})^z$$

$x = 1$, $\quad x+y+z=0$

$$y+z=-1$$

$$-2x-y-2z=0$$

$$y+2z=-2$$

$$z=-1$$

$$y=0$$

$$M = F^1 a^{-1}$$

$\boxed{L}$ $= x=0$, $y+z=1$

$$-y-2z=0$$

$y=z$, $z=-1$, $y=-2z$

$$L = V^2 a^{-1}$$

$\boxed{T}$ $x=0$, $y+z=0$

$$-y-2z=1$$

$y=1$, $z=-1$

$$T = Va^{-1}$$

Zero Correction = - (zero error)

actual value - shown value . shown value - actual value

Que.) T depends on (I, M, g, d)

I → moment of inertia

d → distance

$$T = K [I]^x [M]^y [g]^z [d]^w$$

$$T = [ML^2]^x [M]^y [LT^2]^z [L]^w$$

$x+y=0$, $2x+z+w=0$

$-2z=1$

$z=-1/2$

$x=-y$, $-2y-\frac{1}{2}+w=0$

$w-2y=\frac{1}{2}$

$x=-\frac{1}{2}(w-\frac{1}{2})$

$w=2y+\frac{1}{2}$

$y=\frac{5}{2}(w-1)$

Que) T depends on (I, Mg, d)

$T=K[ML^2]^x[MLT^{-2}]^y[L]^w$

$x+y=0$ $\quad 2x+y+w=0$ $\quad -2y=1$

$x=-y$ $\quad 1-\frac{1}{2}+w=0$ $\quad y=-\frac{1}{2}$

$x=\frac{1}{2}$ $\quad w=-\frac{1}{2}$

$T=K(I)^{1/2}(Mg)^{-1/2}(d)^{-1/2}$

$T=K\sqrt{\frac{I}{Mgd}}$